

Patterns of tree replacement: canopy effects on understory pattern in hemlock - northern hardwood forests*

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Keywords: *Acer saccharum*, Beech-maple forest region, *Betula lutea*, Canopy-understory interaction, Climax, Community stability, *Fagus grandifolia*, Gap phase, Hemlock-northern hardwood forest region, *Tilia americana*, Tree replacement pattern, *Tsuga canadensis*

Abstract

The effect of canopy trees on understory seedling and sapling distribution is examined in near-climax hemlock-northern hardwood forests in order to predict tree replacement patterns and assess compositional stability.

Canopy trees and saplings were mapped in 65 0.1-ha plots in 16 tracts of old-growth forests dominated by *Tsuga canadensis*, *Acer saccharum*, *Fagus grandifolia*, *Tilia americana*, and *Betula lutea* in the northeastern United States. Seedlings were tallied in sub-plots. Canopy influence on individual saplings and sub-plots was calculated, using several indices for canopy species individually and in total. For each species sapling and seedling distributions were compared to those distributions expected if saplings were located independently of canopy influence. Non-random distributions indicated that sapling and seedling establishment or mortality were related to the species of nearby canopy trees. Hemlock canopy trees discriminate against beech and maple saplings while sugar maple canopy favors beech saplings relative to other species. Basswood canopy discourages growth of saplings of other species, but produces basal sprouts. Yellow birch saplings were rarely seen beneath intact canopy.

Since trees in these forests are usually replaced by suppressed seedlings or saplings, canopy-understory interactions should influence replacement probabilities and, ultimately, stand composition. I suggest that hemlock and basswood tend to be self-replacing, maple and beech tend to replace each other, and birch survives as a fugitive by occupying occasional suitable gaps. This suggests that these species may co-exist within stands for long periods with little likelihood of successional elimination of any species. There is some suggestion of geographical variation in these patterns.

Introduction

Species co-existence and biological interactions

Studies of species co-existence may be approached on various spatio-temporal scales. Explanations for the maintenance of floristic richness over whole landscapes rely on analyses of species distribution on coarse-scale successional or environmental gradients. In contrast, understanding of the factors controlling species co-occurrence within stands may require consideration of fine-scale bio-

* Nomenclature follows Fernald (1950).

** Acknowledgements: There are many whose help has contributed to this paper but Robert Whittaker played the primary advisory role throughout. The research described sprang from his ideas and his help was crucial in the performance of the research. I am also grateful to those who helped in the field work and the agencies and individuals who allowed access to field sites. I thank Peter Marks and Margaret Davis for their assistance and suggestions and Robert Peet and Steward Pickett for thorough and helpful review of the manuscript. The research was financed - in part - through a National Science Fellowship and by a McIntyre-Stennis grant.

logical interactions. Although intra-stand variability in the physical environment, such as substrate and microtopography, certainly facilitates niche differentiation and species co-existence (Bratton, 1976a, 1976b; Hicks, 1981), at this scale it is likely that interactions between plants and populations will be relatively more important in controlling community dynamics and composition. It may also be hypothesized that control of species co-existence within stands will pass from external physical factors to internal biological interactions as disturbances shift from severe, where large numbers of dominant individuals are killed, to relatively mild, where death of individual dominants occurs largely independently of the death of others. Biological interactions should also be more strongly asserted as the interval between disturbances lengthens.

The distinction between external, abiotic controlling factors and biological interactions is certainly less than absolute. Broad patterns of climate, substrate, and intra-stand patterns of biological influences are parts of the same continuous spectrum of environmental influences to which all individuals are subjected and which determine vegetation patterns at all scales. The ultimate controlling factors are, in any case, the same; a plant can become established only in microsites which propagules reach and which have adequate light, water, and nutrients.

This paper presents results of a study aimed at discovering biological interactions which could cause co-existence of several species in stands subject, primarily, to small, infrequent disturbances. Observations from a number of stands are used to suggest a pattern of population interactions in a particular range of forest communities.

Canopy-understory interactions in forests

Jones (1945) proposed that forests dominated by shade-tolerant species are generally subject to only mild disturbance in the form of single or few tree gaps since dominance by shade tolerants is likely to be achieved only after long periods without massive disturbance. Jones further argued that these small gaps are usually filled by suppressed saplings of the dominant species. Such replacement may, in the absence of large disturbances over a period of two or more generations of canopy dominants, lead to increased stability in stand species composition.

Two further propositions, added to that of Jones, suggest a way in which co-existence may arise from interactions between individuals. The first proposition is that important niche differences between co-existing plants may be manifested only at certain stages of life-histories, particularly the early stages (Grubb, 1977). In the case of forest trees niche differences allowing co-existence may apply primarily to seedlings and saplings. Changes in niche relationships with age (or size) may involve ontogenetic changes or may simply be due to seedlings and saplings experiencing the environment at a finer scale than mature trees and so experiencing greater variability in microsite conditions. Several workers have discussed the significance of early parts of the life-cycle in understanding the distribution of plants in niche space and in real space and time. The 'microsite mosaic' of Whittaker (1975) and Whittaker & Levin (1977), the 'safe site' of Harper (1977), and the 'regeneration niche' constructed by Grubb (1977) are all similar in their emphasis on the importance of environmental conditions affecting germination, establishment, and early growth.

The second proposition invokes, as a source of variability among microsites, the influence of nearby canopy trees. Canopy trees of different species might differently affect subjacent microsites such as through differences in shade or litter quality, nutrient or water use, or root competition. In this manner canopy individuals could influence seedling and sapling success which, in turn, could lead to co-existence of species through self-replacement, cyclic succession on microsite, or a combination of these patterns.

Effects of canopy trees on understory environment have frequently been implicated as factors controlling successional processes and even the dynamics of relatively stable communities. Whittaker (1975) proposed a microsite mosaic model wherein community stability could be maintained by modification of microsites by canopy trees. Various mechanisms by which such modification can be effected have also been proposed. Connell (1970) and Janzen (1970) have suggested that species-specific herbivores on canopy trees might inhibit growth of conspecific seedlings on subjacent microsites. Alexandre (1977) has shown evidence of such a pattern involving a nematode parasitic on a rain forest tree, but such mechanisms have not yet been documented in temperate forests. Several attempts

have been made to deduce the existence of canopy-understory interactions in temperate forests in the northeastern United States through analysis of distributions of canopy trees, saplings, and seedlings (Forcier, 1975; Fox, 1977; Horn, 1971, 1975). Others (Runkle, 1981; Barden, 1979; Brewer & Merritt, 1978) have attempted to predict replacement patterns by associating sapling species in gaps with tree species which died to form the gaps.

Hemlock-northern hardwood forests

Prior to European settlement mesic uplands in a broad belt from the western Great Lakes region to New England were occupied by forests dominated by shade-tolerant, long-lived trees. The primary dominants in undisturbed stands of these forests are the shade-tolerant sugar maple (*Acer saccharum*), American beech (*Fagus grandifolia*), and eastern hemlock (*Tsuga canadensis*), plus the less tolerant yellow birch (*Betula lutea*) and basswood (*Tilia americana*). The range of forests dominated primarily by these species corresponds roughly to the combined area of the maple-basswood, beech-maple, and hemlock-white pine-northern hard-

woods types of Braun (1950) shown in Figure 1. Composition varies over this range and hemlock-northern hardwood forests grade continuously into others. Beech is most important in the east and south, its range barely reaching the Upper Great Lakes. Hemlock reaches neither the southern nor western limits of the geographical area defined. Yellow birch is most important in the northern and basswood in the western extent of the region. Only sugar maple is an important dominant throughout.

The predominance of shade-tolerant species suggests that these forests may fit the model of a community in which biological interactions are important in controlling community dynamics. Large-scale disturbances in mesic uplands appear to be quite infrequent. None of the five species discussed show resistance or reproductive adaptations to fire. Their total dominance in a stand is evidence that no significant fire has occurred in several centuries. Extensive wind-throw may be the most frequent major disturbance in mesic upland stands of the region, particularly in coastal areas subject to hurricanes. Inland forests are also apparently subject to destruction by strong down-burst winds associated with thunderstorms and by torna-

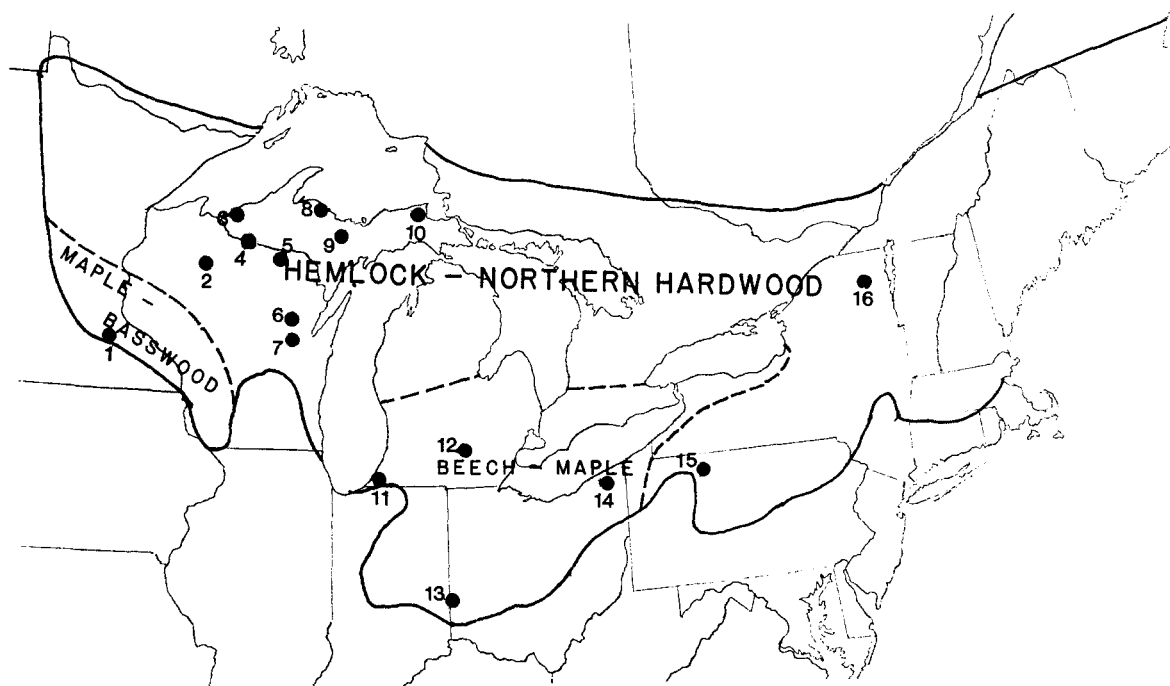


Fig. 1. Distributions of the forest types involved in this study with the beech-maple, maple-basswood, and hemlock-white pine-northern hardwood regions of Braun (1950) delineated. Numbered dots correspond to stands listed in Table 1.

does. However, Canham & Loucks (1983) have determined that, in northern Wisconsin, such disturbances have a return time in excess of 1 000 years – three to five times the period generally spent in the canopy by individuals of the species under study. Small canopy gaps associated with deaths of individual trees are apparently the chief mode of canopy disruption for relatively long periods. Runkle (1981), Barden (1979), Fox (1977), and Brewer & Merritt (1978) have shown that these gaps are usually filled by individuals of the dominant species as predicted by Jones. Forest on less mesic sites or in large areas without natural firebreaks such as the western Great Lakes region are subject to more frequent fire and their composition reflects this.

Methods

Field methods

Stands to be sampled were required to be 1) strongly dominated by some combination of the five major species listed, 2) free of evidence of disturbance other than gaps caused by natural death of one or a few trees, and 3) fairly homogeneous in substrate and topography. These criteria should best permit isolation of effects of canopy trees on the distribution of seedlings and saplings. No contention is made that such stands were generally abundant in primeval forests, but these effects should be present in all forests, though they may be strongly confounded in more disturbed stands. Nine hemlock–hardwood stands were sampled in northern Michigan and Wisconsin, and two in Pennsylvania and New York. Four beech–maple stands were sampled in southern Michigan and Ohio, and one maple–basswood forest in southern Minnesota (Fig. 1, Table 1). Sites are concentrated in northern Wisconsin and Michigan due to difficulties in satisfying the second and third criteria for stand selection in areas of greater topographic diversity and more pronounced anthropogenic disturbance.

Two to eight 0.1-ha (20×50 m) quadrats were placed in each stand. This usually involved pacing a predetermined distance along a transect or trail, then moving a predetermined distance in a predetermined direction from the point reached. In each quadrat saplings (individuals >2 m tall and <15 cm

Table 1. Names and locations of sample sites, with sample numbers.

Site number	Name	Location	Sample numbers
1	Nerstrand Woods State Park	Rice Co., Minn. T110N, R19W	52–54
2	Flambeau 'Big Block' Scientific Area	Flambeau St. For. Sawyer Co., Wisc. T37N, R3W	12–16
3	Porcupine Mts. State Park	Ontonagon Co., Mich. T50N, R44W	7–11
4	Sylvania Tract	Ottawa Nat. For. Gogebic Co., Mich. T44N, R40W	1–6 42–43
5	Bose Lake Scientific Area	Nicolet Nat. For. Forest Co., Wisc. T40N, R12E	17–20, 44
6	Jung Beech–Hemlock Scientific Area	Shawano Co., Wisc. T27N, R14E	37–41
7	Tellock's Hill Scientific Area	Waupaca Co., Wisc. T24N, R13E	35–36
8	Huron Mountain Club	Marquette Co., Mich. T52N, R28W	45–49
9	Dukes Experimental Forest	Hiawatha Nat. For. Marquette Co., Mich. T46N, R23W	26–29
10	Tahquamenon Falls State Park	Luce Co., Mich. T49N, R8W	21–25
11	Warren Dunes State Park	Berrien Co., Mich. T7S, R20W	55–60
12	Irwin's Woods	Jackson Co., Mich. T2S, R2E	63
13	Hueston Woods State Park	Preble Co., Ohio	61–62
14	Hiram College Biological Station	Portage Co., Ohio	64–65
15	Tionesta Tract	Allegheny Nat. For. McKean Co., Pa.	30–34
16	Saranac Lake	Adirondack State Park, N.Y.	50–51

diameter breast height) and canopy trees (>15 cm dbh) were recorded by species and diameter and mapped (canopy trees outside quadrats were included in the sample if they were within 8 m of saplings within the quadrat). Sapling and canopy stems were mapped using tapes and an optical range-finder to within 1 m of their true location (accuracy was generally greater). In each 0.1-ha quadrat 25 or 50 1-square meter sub-plots were placed along the long axis (in later quadrats only every other sub-plot was sampled). In each of these sub-plots, seedlings of all woody species were counted and divided into 3 height classes: less than 20 cm, between 20 cm and 1 m, and between 1 m and 2 m.

Analytic methods

Previous attempts to describe the effects of canopy trees on understory patterns have used more or less subjective, qualitative evaluations of canopy influence. Most have involved identification of the single canopy species most dominant at a point on the forest floor. In practice this usually has meant the species of the canopy individual nearest a seedling or sapling or the species of the tree which died to form a gap. An objective in this study was to quantify canopy influence by incorporating the size and distance from an understory point of all canopy trees within a set distance. Canopy influence, whether through shading, allelopathy, competition in the rhizosphere, or any other mechanism, should be in some way additive and dependent on tree size and spacing.

Attempts have been made to quantify competitive interactions among trees. Previously developed indices (e.g. Bella, 1971; Alemdag, 1978; Lorimer, 1981) were not used here because they either do not quantify influence in the way specified above, are not appropriate for dealing with individual microsites, deal with species not involved in this study, or are based on models of early successional communities. Three indices were devised for this study each based on a different model of the decline in influence of a particular canopy tree with increasing distance (Fig. 2). I_L assumes influence to decrease linearly with distance, I_N is represented by a bell-

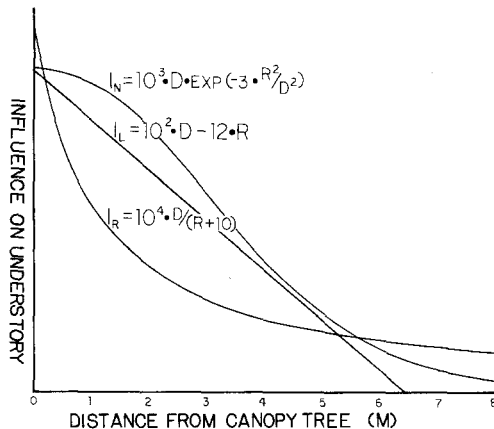


Fig. 2. Graphical representation with formulae for models of variation of canopy influence with distance from canopy tree. All curves are calculated for a canopy tree of 40 cm diameter. In formulae D is diameter of canopy tree, and R is distance from tree.

shaped curve and assumes a plateau of high influence extending some distance from the tree, and I_R assumes influence to vary with the reciprocal of distance (the 'tails' on I_R and I_N which are lacking on I_L proved to affect results little). These curves were subjectively drawn to imitate a range of biological models. Influence of individual trees as expressed by these indices may be calculated and added for any number of trees.

Values were calculated by each of the above indices for influence of individual canopy species and for total canopy influence on each sapling within the 0.1-ha quadrats and on each square-meter seedling quadrat. Only trees within 8 m of a sapling or sub-plot were used in calculating indices; this distance nearly always allowed inclusion of any tree whose canopy extended above the sapling and usually included at least three or four canopy trees. Similar values were obtained using a more conservative, non-distance-weighted index which was simply the sum of basal areas of canopy trees within the 8 m limit.

An expectation for these indices under the null hypothesis that seedlings or saplings are distributed independently of canopy tree influence may be calculated if density and diameter distributions are known for each canopy species. The formula for this expectation as derived in Woods & Whittaker (1981) is

$$E(I) = \sum_{i=1}^n p_i 2\pi \int_0^x R f(R) dR$$

where the summation is over size classes i to n , p_i is density in size class i , x is the maximum distance for inclusion of trees (here 8 m), and $f(R)$ is the index of canopy influence as a function of distance (with diameter held constant).

Patterns of relationships between canopy and sapling distributions were then examined by comparing observed values of indices to corresponding calculated expectations (Woods & Whittaker, 1981). Comparisons were made of influence of individual canopy species and of total canopy influence.

In another analysis (see Woods, 1979) pairwise comparisons of sapling species responses to indices were formed by taking the difference between indices of the influence of two canopy species.

$$I_n = I_A - I_B$$

I_A and I_B are influence of species A and species B. I_n , or net influence will be zero if influence of the two species is equal, positive if species A is more important ($I_A > I_B$), and negative if species B is more important ($I_B > I_A$). Pairs of sapling species were compared in their response to net canopy influence of pairs of canopy species by ranking all saplings of both species by values of a particular I_n and comparing rankings of the two sapling species with a Wilcoxon rank-sum test. Significant results allow rejection of the null hypothesis that the two sapling species are distributed similarly with respect to the joint influence (I_n) of the two canopy species involved. Rejection means that one or both sapling species is relatively more abundant near trees of one

of the two canopy species. Similar tests for all pairs of sapling species for all possible I_n 's permitted comparison of the relative responses of particular sapling species to particular canopy species.

Indices of canopy influence on seedling sub-plots were tested for correlation with number of seedlings in each species and height class.

For all of these treatments the 0.1-ha quadrats were grouped into twelve composite samples. For results to be meaningful, the analyses described need to be performed only on groups of samples of similar composition. Nine such composite samples were formed by pooling samples closely associated in a detrended correspondence analysis (DECORANA; Hill & Gauch, 1980) ordination (Fig. 3).

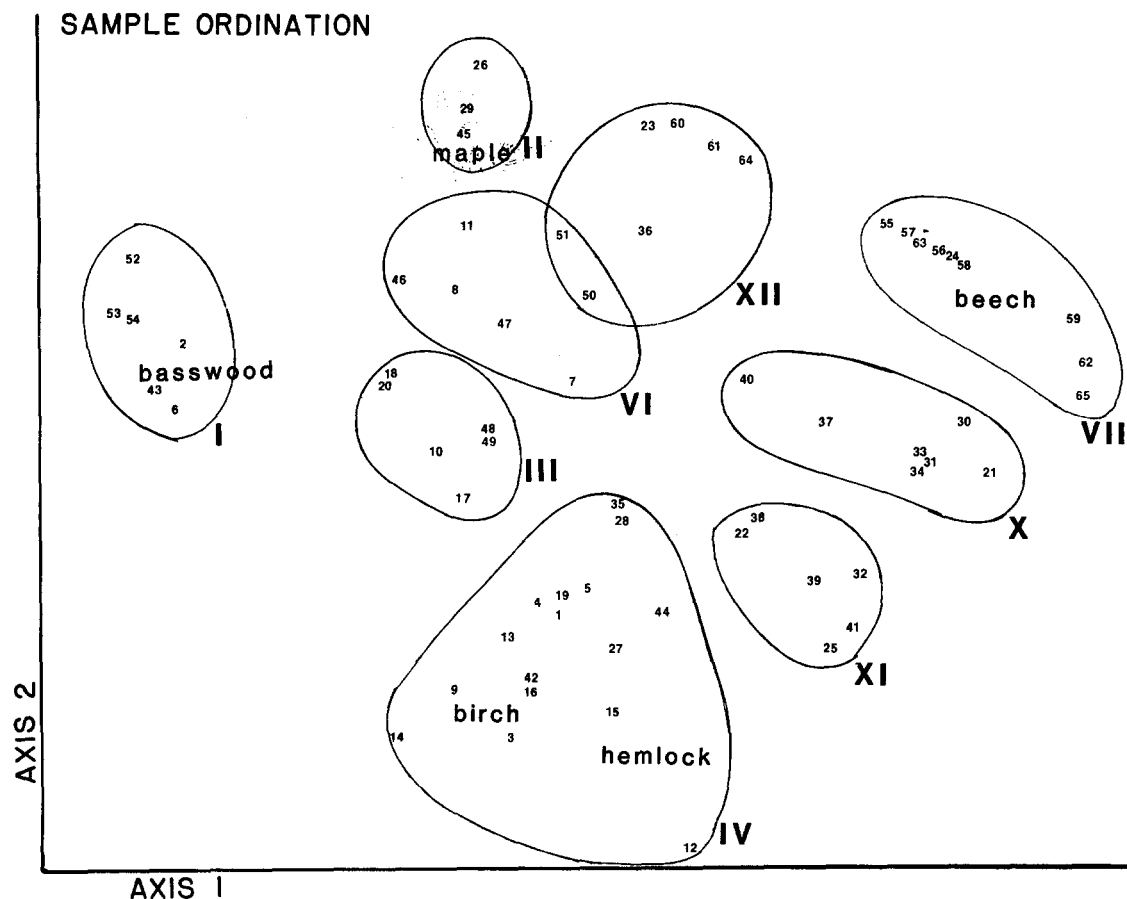


Fig. 3. Distribution of all samples from stands shown in Figure 1 on the first two axes of DECORANA ordination. Groups are composite samples. Areas of maximum abundance are indicated for each species (beech at the right of the ordination, basswood at the left, and hemlock and yellow birch at lower center). Maple is significant throughout the ordination but shows maximum abundance at upper center. Composites 5, 8, and 9 are not illustrated, but are composed of quadrats from the Huron Mountains (45–49), Jung (37–41), and Tionesta (30–34) stands respectively.

These groups were formed by pooling nearest quadrats or groups with nearness determined by distance between the nearest pair of quadrats in the two groups (single linkage classification, sensu Pielou, 1977). Composites 6 and 12 overlap because quadrat 50 is equidistant from 36 and 47. Three other composite samples (5, 8, and 9) were made by pooling quadrats from relatively uniform stands. It was hoped that these would reveal any geographical changes in patterns of interaction that would be obscured by forming composites from quadrats from different stands. Composite samples will hereafter be referred to as samples. The samples are not all independent as some quadrats are placed in more than one sample (Fig. 3). Other groupings into composite samples were tried and yielded similar results (results in Woods & Whittaker, 1981, are from a different grouping).

Results of these tests were compared with those of a simpler set of tests. For each sapling species tests for correlation of sapling abundance with basal area of the various canopy species were performed using Spearman's rank-correlation coefficient. Coefficients were calculated using whole 0.1-ha quadrats as samples and on sample sets obtained by dividing large quadrats into two 20×25 m plots and into four 10×25 m plots. Each correlation of sapling abundance and canopy basal area was calculated using only samples from stands within the geographic range of both species involved in the test.

All tests of association between canopy trees and sapling distributions were done with and without inclusion of beech root sprouts.

Results

Basal areas for the composite samples are listed by species in Table 2. Total basal area ranged from 35 to 56 m^2/ha , densities of canopy trees from 200–500 stems per ha and sapling densities from 100–4 000 stems per ha.

Saplings and seedlings of the dominant species showed extreme variation in abundance. At least 500 saplings of each of the shade-tolerant species were involved in the study. Maple saplings were three times as abundant as beech and hemlock saplings were somewhat rarer (beech root suckers are not included in this total, but represented only

Table 2. Distribution of basal area by species in composite samples.

Sample number	Species basal area (m^2/ha)						Sum
	<i>Acer</i>	<i>Betula</i>	<i>Fagus</i>	<i>Tilia</i>	<i>Tsuga</i>	Other ^a	
1	14.4	2.4	0.0	20.7	0.5	6.9	44.9
2	36.3	3.0	0.0	3.1	3.8	0.2	46.4
3	18.8	7.9	0.0	8.3	18.2	0.4	53.6
4	7.2	10.9	0.5	2.3	30.3	5.1	56.3
5	25.0	3.2	0.0	6.0	12.0	0.4	46.6
6	29.8	3.6	0.3	2.9	11.9	0.4	48.9
7	8.2	0.0	25.7	0.0	0.0	3.0	36.9
8	8.8	1.4	16.6	0.0	25.8	0.0	52.7
9	2.2	1.1	19.2	0.0	12.0	0.9	35.6
10	5.6	1.0	18.9	0.0	10.8	0.7	37.0
11	3.7	2.5	13.1	0.0	24.5	1.8	45.5
12	24.2	0.5	7.1	0.2	4.4	1.3	37.7

^a Other species include *Abies balsamea*, *Acer rubrum*, *Acer pensylvanicum*, *Betula lenta*, *Carpinus caroliniana*, *Carya cordiformis*, *Fraxinus americana*, *Fraxinus nigra*, *Juglans cinerea*, *Liriodendron tulipifera*, *Ostrya virginiana*, *Picea rubens*, *Pinus strobus*, *Prunus serotina*, *Quercus borealis*, *Thuja occidentalis*, *Ulmus americana*, and *Ulmus rubra*.

about 10% of beech saplings). Only 40 yellow birch saplings were found, mostly in small canopy gaps. Basswood basal sprouts were abundant but only 16 independent saplings were found in the samples. Seedling abundances showed similar variation among species.

Comparison of observed and expected canopy influence

Comparisons of observed and expected canopy influence gave similar results for each of the indices including the non-distance-weighted basal area measure. There were slight differences due to peculiarities of the models (Fig. 2) but such differences did not significantly affect the results. The results displayed below were obtained using I_R with exclusion of beech root suckers. Differences observed when suckers were included were small and are discussed separately.

Comparisons of observed and expected hemlock canopy influence for all sapling species are displayed in Figure 4. This figure and Figures 5 and 6 are similar to those presented in Woods & Whittaker (1981) but illustrate results obtained from an improved grouping into composite samples. In this and similar figures each row of symbols represents a

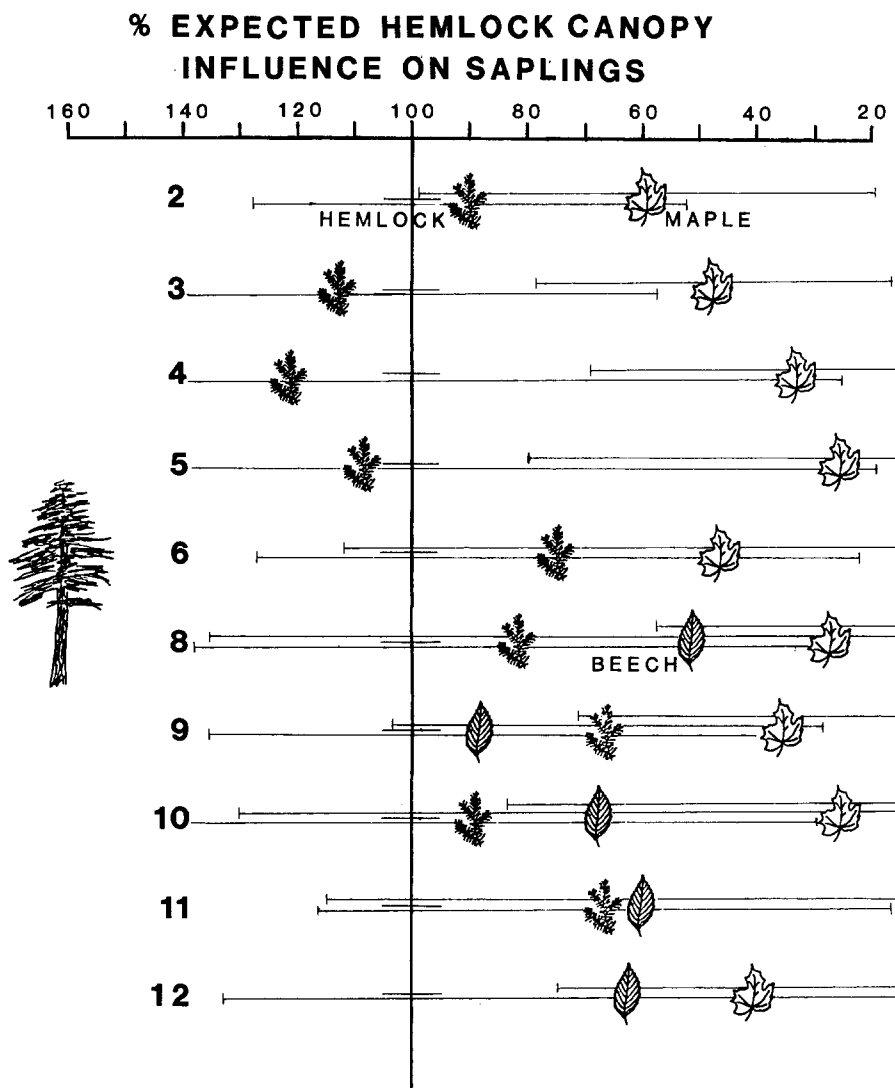


Fig. 4. Distribution of saplings by species with respect to hemlock canopy influence. Leaf symbols represent sapling species, rows single composite samples (sample numbers given at left). The horizontal axis (top) is hemlock canopy influence as percent of its calculated expectation and is thus independently scaled for each sample. Leaf symbols are located at mean values of relative hemlock canopy influence observed for all saplings of that species in that sample. Sapling species for which symbols are located further to the right in the diagram are found, on the average, at lower levels of hemlock canopy influence. Error bars associated with each sapling symbol indicate one standard deviation (error bars are not shown in succeeding figures, but those shown here are representative). For each sample, the bottom error bar corresponds to the sapling species at highest relative canopy influence. Only samples with significant representation of hemlock in the canopy are shown.

single composite sample. Only samples in which the canopy dominant (hemlock in this case) constitutes >10% total basal area are shown.

The null hypothesis level of hemlock canopy influence was exceeded on the average only by hemlock saplings, and hemlock saplings were, in eight of nine instances, found at higher hemlock canopy

influence than any other species (in the exception, Tionesta, beech was at higher hemlock influence). This consistency of ranking permits rejection at $p < 0.05$ of the null hypothesis of similar sapling response, even if 'overlapping' samples 5, 8, and 9 are not considered. In nine of nine samples maple saplings were found at the lowest average hemlock

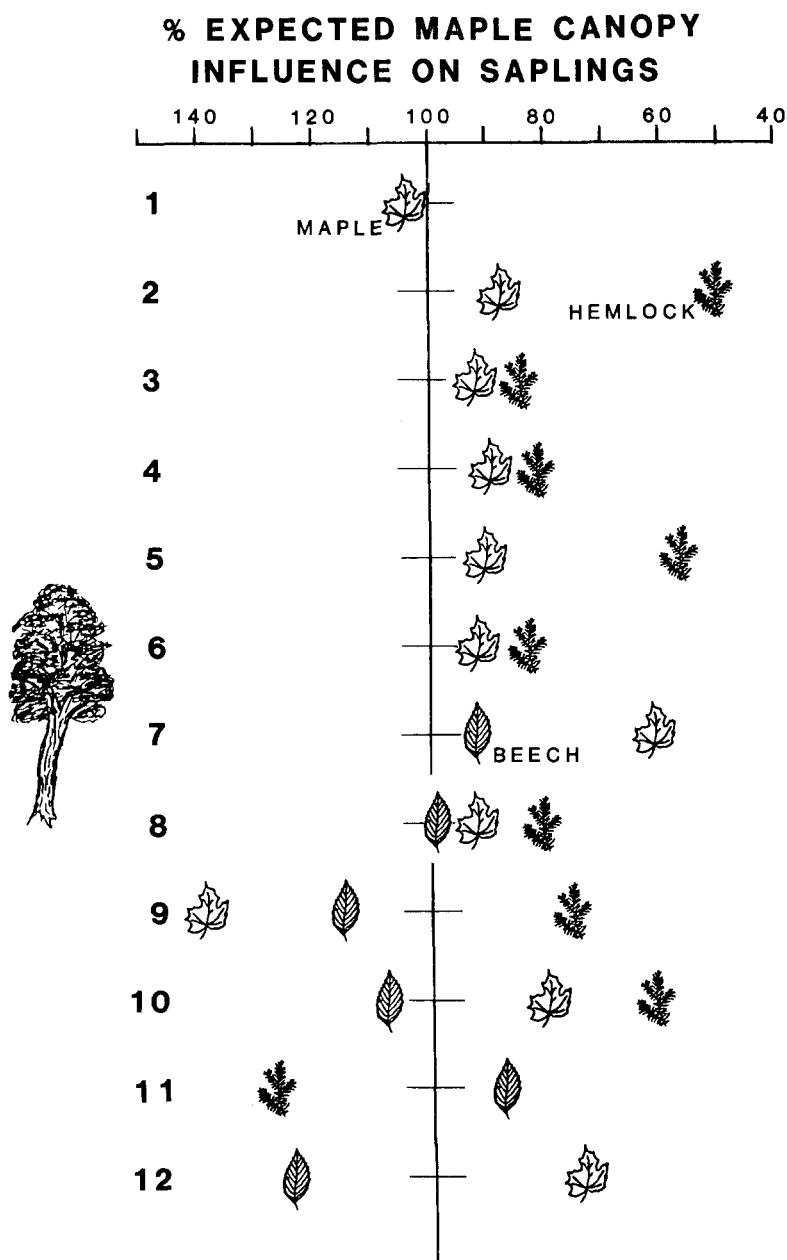


Fig. 5. Distribution of saplings by species with respect to sugar maple canopy influence. Representation is as described for Figure 4.

canopy influence of all species (this consistency allows rejection of the same null hypothesis, with the same considerations, at $p < 0.05$). In only one case did the mean for maple saplings fall within one standard deviation of the expected value. Finally, beech saplings were found, on average, at higher hemlock influence than were maple saplings, though

not as high as would be expected by chance. Avoidance of hemlock canopy was shown by maple saplings and, to a lesser degree, beech saplings, while hemlock saplings showed no clear trend.

The organizing effect of maple canopy influence (Fig. 5) on the understory was apparently not as great as that of hemlock. Maple saplings tended to

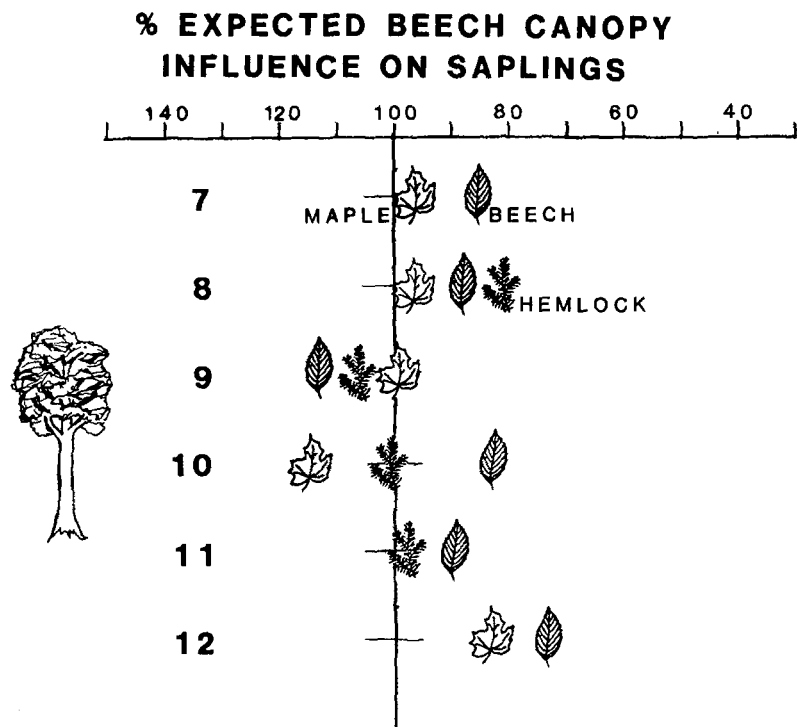


Fig. 6. Distribution of saplings by species with respect to beech canopy influence. Representation as in previous figures.

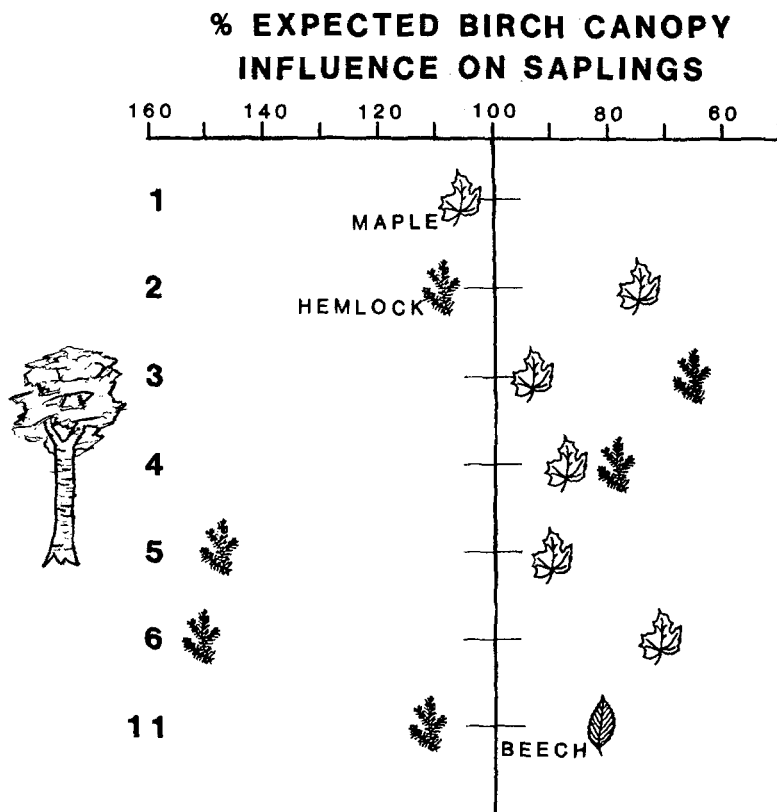


Fig. 7. Distribution of saplings by species with respect to yellow birch canopy influence. Representation as in previous figures.

be found near but slightly below expected levels of maple influence. In four of five cases, beech saplings were at higher maple influence than were maple saplings (consistency of ranking significant at $p=0.05$). The most notable response was by hemlock saplings where in eight of nine cases they were found below expectation and at lower maple influence than any other sapling species (the exception is again Tionesta).

Beech canopy (Fig. 6) and yellow birch canopy (Fig. 7) elicited little clear response in sapling distribution, mean influence values for all sapling species being scattered about the expected values. Possible tendencies for beech saplings to be found at lower beech canopy influence than maple saplings (Tionesta again an exception) and hemlock saplings to be found at high birch canopy influence are consistent with patterns discussed below.

Saplings of maple and hemlock were apparently inhibited by basswood canopy influence (Fig. 8). However, the basswood saplings were almost exclusively basal sprouts of canopy trees and so were found at extremely high levels of basswood canopy influence.

The plot of sapling distributions with respect to total canopy influence (Fig. 9) suggests that maple saplings are found at lower canopy influence than saplings of the other shade-tolerant dominants. Beech and hemlock saplings also generally occur at lower than expected values of canopy influence, though average values for beech saplings tend to be higher than those for hemlock. All sapling species were found at relatively low canopy influence, a not surprising confirmation that saplings are most abundant in gaps or beneath light canopy.

Figure 10 shows average canopy influence by various species on beech saplings. The ranking of shade-tolerant canopy trees by their mean influence on beech saplings is highly consistent; the ratio of observed to expected influence of maple canopy on beech saplings was higher than that of beech canopy in all but one sample (sample 9). In short, beech saplings show less avoidance of maple canopy than beech canopy. Similar analyses were conducted for maple and hemlock saplings with results consistent with those shown in Figures 4–9; hemlock saplings were found at high relative values for conspecific influence and low values of maple influence while

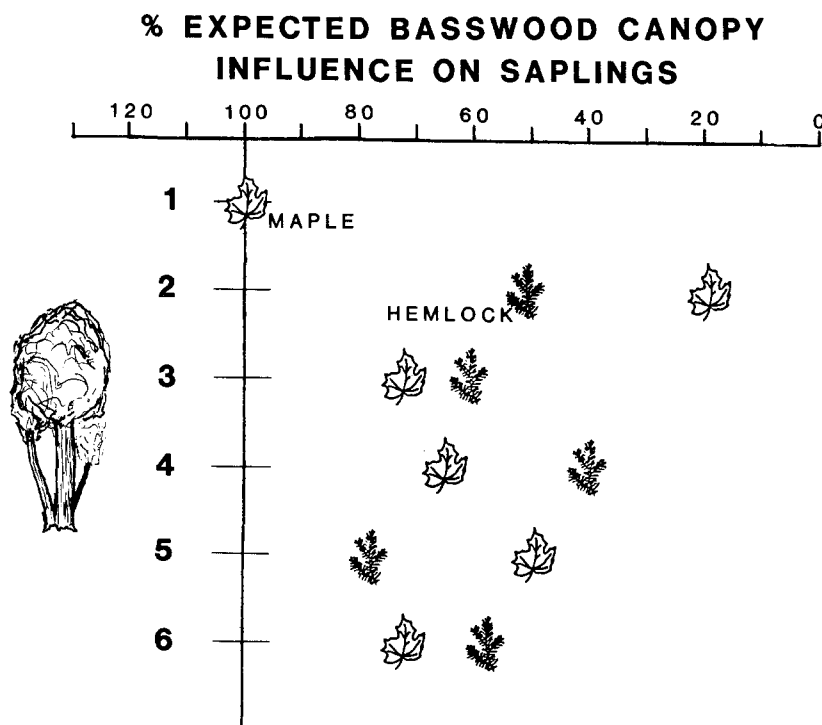


Fig. 8. Distribution of saplings by species with respect to basswood canopy influence. Representation as in previous figures.

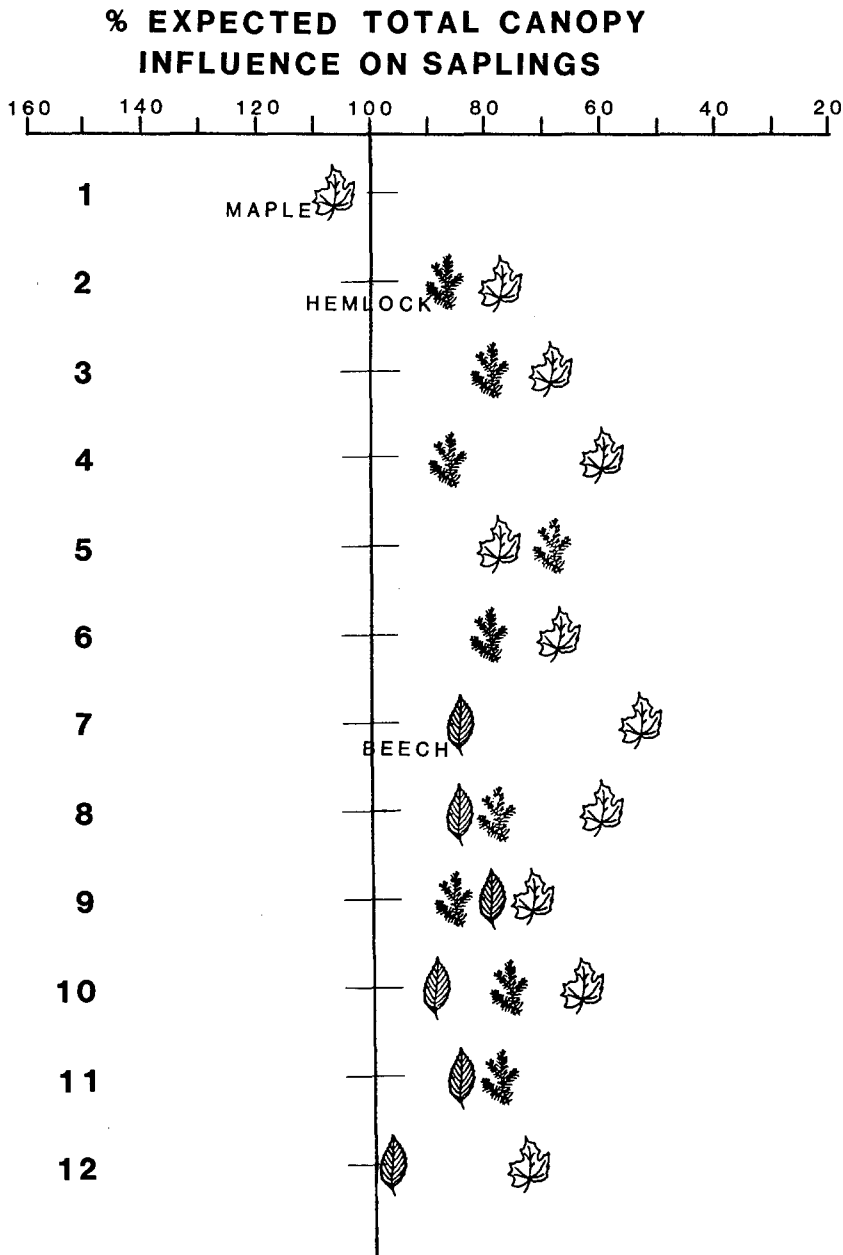


Fig. 9. Distribution of saplings by species with respect to total canopy influence of all species. Representation as in previous figures.

maple saplings were always at much lower values of hemlock canopy influence than would be expected by chance.

Similar analyses of dead sapling distributions were compared with these for live saplings to identify differential mortality caused by differences in canopy influence. Dead maple saplings showed distributions similar to those of live saplings. Both

beech and hemlock saplings showed increased relative abundance of living saplings at high maple canopy influence. Living yellow birch saplings tended to be found at low canopy influence of all species, while higher canopy influence was associated with greater relative abundance of dead yellow birch saplings.

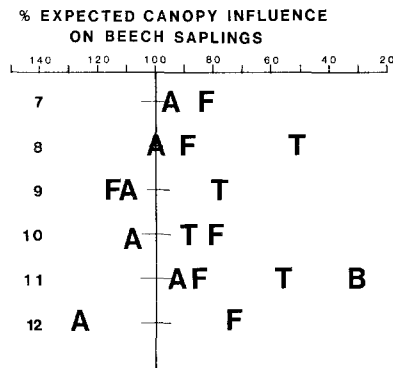


Fig. 10. Distribution of canopy species with respect to their average influence on beech saplings. Representation is similar to that in previous figures with horizontal lines representing canopy influence as percent of expectation, vertical line representing expectation, and rows of symbols corresponding to samples. In each sample symbols representing canopy species are located according to average influence, relative to expectation, of that canopy species on beech saplings. Symbols are: A = sugar maple, F = beech, T = hemlock, and B = yellow birch. Thus, canopy symbols at right of diagram indicate low relative influence on beech saplings. Only samples with more than 10 beech saplings are shown.

Rank-sum comparisons of net influence

Results of the rank-sum tests (Fig. 11) were consistent with the canopy influence comparisons discussed above. In all tests involving hemlock canopy influence (5, 6, 8, and 9 in Fig. 11), hemlock saplings occurred at greater hemlock influence than saplings of the other species. This strengthens the suggestion of relative avoidance of hemlock canopy by deciduous species. In the absence of hemlock, beech and maple saplings were found at relatively lower influence of conspecific canopy trees than of the other canopy species in the comparison (significant in all but one case). This is most apparent in the maple-beech comparison (1) where this mutual self-avoidance was shown to depart with high significance from random expectation.

Correlations of sapling abundance with canopy basal area

Correlations of sapling abundance with canopy basal area (Fig. 12) further establish hemlock's suppression of saplings and the reciprocal relationship between beech and maple. The null hypothesis

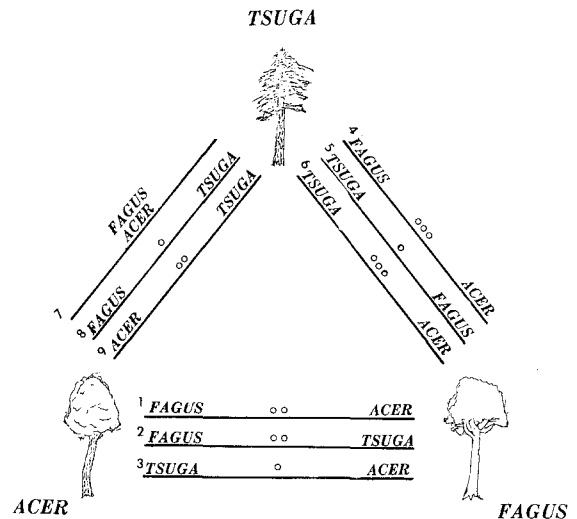


Fig. 11. Pairwise comparisons of response of sapling species to composite indices of canopy influence. Tree symbols at corners of diagram represent canopy influence of indicated species. Each line between tree species represents tests done on one of the possible pairwise comparisons of sapling species response to net influence of the two canopy species connected by the line. All lines represent several comparisons connected by the line. Where rank-sum tests indicated significant differences in response of saplings species, saplings species names are shown nearest the canopy species which they, relatively, favored. Levels of significance at which the null hypothesis of similar distribution of sapling species were rejected are indicated by the number of dots above each line: one dot indicated rejection in at least 3 of 4 tests at $\alpha = 0.05$, two dots rejection in all tests at α between 0.005 and 0.05, three dots rejection in all tests with α less than 0.005.

is that canopy trees do not effect distribution of understory individuals. Expected correlations between sapling abundance and conspecific canopy basal area are positive for species with limited dispersal and patchy seed sources, and zero for species with uniformly dispersed seeds.

For all plot sizes hemlock saplings showed positive correlations with conspecific basal area and negative correlations with basal area of both maple and basswood. These become significant at small plot size. Maple saplings were positively correlated with beech canopy and negatively associated with total basal area at all plot sizes. Negative correlations of maple sapling abundance with hemlock basal area were consistent and approached significance at smaller plot sizes. Beech saplings had no significant correlations though they did show positive association with maple basal area and negative

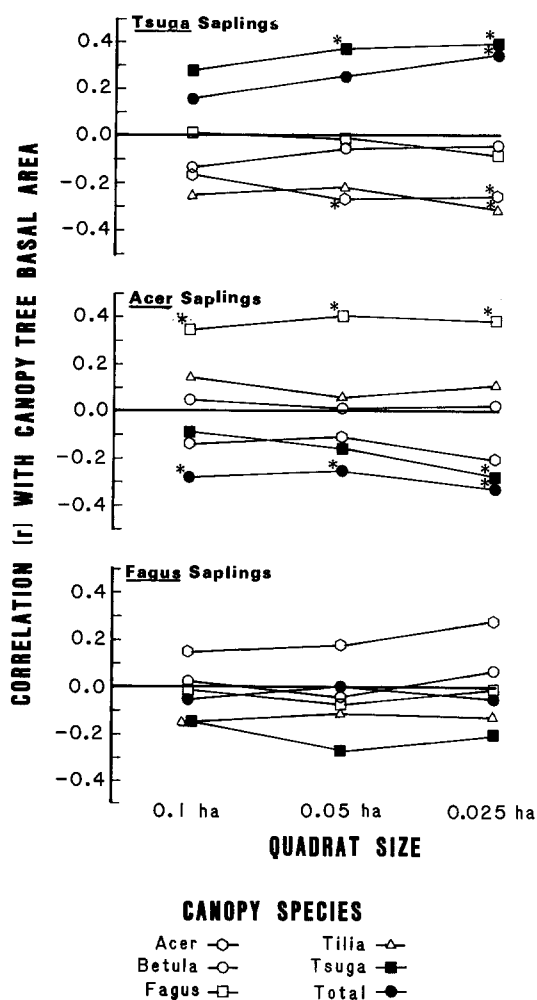


Fig. 12. Correlations of sapling abundance with canopy tree basal area by species of canopy tree and sapling. Each graph shows results involving one of the shade-tolerant sapling species for three plot sizes. Lines connect graphed coefficients for same canopy species. Stars indicate significance of correlation coefficient at $\alpha = 0.05$.

association with hemlock and basswood (the last based on a very small number of co-occurrences).

In nearly all cases particular correlations became stronger with smaller plot size. This was especially notable for hemlock basal area.

Correlations of seedling abundance with canopy influence

Maple seedlings showed strong positive correlation with conspecific influence in the smallest size

classes, but this was lost in the larger classes (Table 3). Correlations with total canopy influence were initially positive but became strongly negative. Maple seedlings were, throughout, negatively associated with hemlock canopy. Similar trends were seen for beech seedlings in correlations with conspecific and total canopy influence, and beech seedlings also showed increasingly negative correlation with hemlock canopy influence in larger seedling size classes. Hemlock seedlings were comparatively rare in the study but may have been positively correlated with conspecific canopy.

Discussion

Canopy influence on understory pattern and tree replacement

The distribution of seedlings and saplings in forest understories may be controlled by fine-scale physical variation, by seed dispersal, or by other canopy tree influences. Samples in this study were chosen to minimize microsite variation. The effect of seed dispersal alone should range from distinct 'seed shadows' for poorly dispersed species to little at all in species with well-dispersed seeds. Consequently, negative association of seedlings and saplings with conspecific canopy trees can be interpreted as a result of canopy influence on reproductive success.

Results show that saplings and seedlings are not distributed through the stand independently of the influence of canopy trees. In some cases association with conspecific canopy is shown, as would be expected if distributions resulted from seed shadows. In other cases saplings were relatively under-represented near conspecific canopy trees. These distributions of seedlings and saplings, with the assumption that saplings are likely successors of current canopy trees, suggest that the probabilities of various species replacing a canopy tree are affected by the species of that tree. Observed distributions may be used to suggest a complex of likely canopy transitions with varying probabilities of occurrence.

Other workers have suggested that canopy influence plays an important role in determining patterns in beech-maple forests. Horn (1975), Whitaker & Levin (1977), and Fox (1977) have described beech-maple forests in which saplings have greater

Table 3. Correlations of seedling abundance with canopy influence. Number of tests is number of samples with more than 25 seedlings of appropriate species and size. Positive correlations are indicated by * ($p=0.01$) or + ($p=0.05$), negative correlations by - ($p=0.01$) or - ($p=0.05$). Only significant correlations are indicated. Size classes are: 1, first year seedlings with cotyledons; 2, seedlings less than 20 cm tall; 3, seedlings between 20 cm and 1 m tall; and 4, seedlings between 1 m and 2 m tall.

No. of tests	Seedling species	Size class	Canopy influence					
			<i>Acer</i>	<i>Betula</i>	<i>Fagus</i>	<i>Tilia</i>	<i>Tsuga</i>	Total
9	<i>Acer</i>	1	**** +++	*		*++ =	====	* ++
9		2	**** ==	** =		*** =	+ - =====	* ==
7		3	** +	* =		** =	==	+ =====
3		4		+		*	-	-
3	<i>Betula</i>	1	=					=
3	<i>Fagus</i>	1	*	* -	*+		*	* -
2		2	=	*+			**+	* -
4		3	-		*		-	-
2		4					=	=
1	<i>Tsuga</i>	2					*	

relative abundance beneath canopy of the other species. In a previous study I have shown (Woods, 1979) sapling frequency distributions of beech and maple to be relatively skewed toward more frequent occurrence beneath the canopy of the other species. Figures 5, 6, 10, and 11 suggest that the interactions between beech and maple described for the special case of strong two-species dominance have some generality in more complex stands. Dead beech saplings showed particular association with high beech canopy influence. In 75% of the samples dead beech saplings were found at higher average beech canopy influence than were living beech saplings. This could be due to greater frequency of establishment beneath parent trees and thus more saplings available to die, but it is also consistent with

the hypothesis of high differential mortality of beech beneath conspecific canopy. The relative abundance of small dead saplings beneath conspecific canopy trees suggests that differential mortality may occur just before or after entry into the sapling size class. Gradual attrition of maple seedlings at high maple canopy influence may be indicated, especially for larger seedling size classes, by data in Table 3.

The causes of the differential success of saplings and seedlings are unknown. Root competition might be especially significant for a seedling or sapling beneath a conspecific canopy tree. The 'natural enemies' hypothesis of Janzen (1970) and Connell (1970) could cause similar patterns. Horn (1971) proposed that differences in shade tolerance

and canopy density could permit co-existence. If maple is more shade-tolerant in the understory and casts relatively light shade in the canopy, while beech canopy casts denser shade and grows more rapidly as saplings in higher light, each may prosper beneath the canopy of the other. Poulson (personal communication) has used detailed work on extension growth of seedlings and saplings in a Michigan beech-maple forest to suggest a mechanism essentially the reverse of Horn's; beech is more shade-tolerant (grows more rapidly beneath intact canopy) while maple, although barely able to survive as suppressed seedlings under full canopy, can grow much more rapidly than beech at high light levels. Poulson claims that maple canopy inhibits both species but beech may survive as saplings while the less tolerant maple does not survive beyond seedling size, with the result that maple trees may be more frequently replaced by beech. Beech canopy, however, inhibits growth of both species in the understory thus confining most regeneration to canopy gaps where more rapidly growing maple is usually more successful. Thus distributions of beech and maple saplings suggest replacement of each species in the canopy by the other, a phenomenon termed reciprocal replacement by Woods (1979). This pattern is supported by studies of forest gaps by Runkle (1981). In order to calculate transition probabilities for species pairs Runkle recorded dominant saplings in small gaps in old-growth forests, assuming these would be most likely to fill the gaps. Transition probabilities calculated by Runkle for Hueston Woods, a beech-maple stand also sampled in this study, indicate a strong tendency toward reciprocal replacement.

Conclusions regarding beech replacement rest, in part, on the exclusion of beech root sprouts from the analysis. When sprouts were included in the rank-sum tests the null hypothesis that maple and beech saplings were distributed similarly with respect to the index of joint canopy influence was still rejected, but at $p = 0.05$ as compared to $p = 0.001$ when sprouts were excluded. Other test results were unchanged. Root sprouts are generally very near the parent tree and so strongly influence the distance-weighted indices. The contribution of root sprouts to the establishment of new canopy beech trees remains unclear and probably varies geographically (Ward, 1961). There is a long-standing debate concerning the ability of root sprouts to develop

into canopy trees (Fowells, 1965). Claims have been made that beech sprouts, particularly those near the parent tree, rarely survive beyond the sapling stage. This may be due to transmission of pathogens from the parent tree (Campbell, 1938) or lack of development of an independent root system (Poulson, personal communication).

Beech and sugar maple were the only shade-tolerant species in the stands studied to exhibit nearly complete dominance, so interactions of other species must be studied in a more complex context.

Hemlock canopy seems to exert the strongest influence on the seedling and sapling strata, apparently inhibiting beech and, especially, maple saplings relative to hemlock saplings. This is unlikely to result from poor seed dispersal on the part of beech and maple since neither species was found at particularly high abundance beneath conspecific canopy compared to other species than hemlock. Beech seedlings were positively associated with hemlock canopy (Table 3) and maple seeds are so widely dispersed that dispersal limitation seems improbable. The most likely explanation is higher mortality of beech and maple seedlings beneath hemlock canopy. Beech seedlings may not experience this mortality in their smallest sizes while maple seedlings are apparently affected very soon after establishment. Dead saplings of beech and maple were not particularly common beneath hemlock suggesting earlier mortality. Dead hemlock saplings, on the other hand, showed increased frequency at high hemlock canopy influence suggesting a gradual attrition rather than a strong limitation at early stages.

The abundance of hemlock saplings beneath conspecific canopy is only relative; they were relatively few in number. Once established, however, hemlocks typically show greater shade tolerance. Increment cores of saplings showed as many as four distinct episodes of suppression. Dead hemlock saplings were comparatively rare. The association between hemlock saplings and canopy becomes more pronounced at finer scales (Fig. 12). Hemlock canopy trees were shown by Peterson's (1976) index of segregation to exhibit strong positive contagion which should intensify the canopy-understory interaction.

Hemlock canopy has been suggested to have a negative influence on understory species in Indiana (Daubenmire, 1936), central New York (Lewin,

1974), hemlock's northern range (Rogers, 1980), and in the Great Smoky Mountains (Hicks, 1981). In this study, total sapling density was significantly lower in quadrats with high hemlock dominance. Suggested mechanisms for this suppressing effect of hemlock include shade effects of its dense evergreen canopy and allelochemical or mechanical inhibition by hemlock litter. Taken together, these patterns suggest that hemlock is most frequently self-replacing.

Maple canopy has effects on the understory beyond the relative success of beech over maple saplings. Hemlock saplings in nearly all cases were disfavored by maple canopy. Again, this pattern could be a result of poor dispersal of seed from the locality of hemlock parent trees, but this is not supported by the distribution of hemlock saplings with respect to other canopy species. Mechanisms which might cause enhanced mortality of hemlock beneath maple canopy are not obvious. Tubbs (1973, 1976) did find allelopathic suppression of several species of conifers by root exudates of maple seedlings, but hemlock was not one of the species tested. Maple seedlings in areas with maple dominance usually form a dense understory stratum magnifying the potential for allelopathic effects.

Yellow birch is a 'gap-phase' species, not tolerant of heavy shade (Fowells, 1965) but capable of rapid growth and exploitation of relatively small canopy gaps where light levels are moderate to high (Runkle, 1981; Barden, 1979; Tubbs, 1969; Trimble, 1970; Forcier, 1975). In this study, since large gaps were intentionally unsampled, saplings of birch were relatively rare and could usually be associated with some recent canopy opening. The few birch saplings were found at much lower levels of total canopy influence than expected. If gaps are of great enough frequency, yellow birch should be able to persist as an important canopy species, though its importance may, to some extent, reflect previous severe disturbances. Hemlock and yellow birch were strongly associated in the canopy and showed similar distributions in ordinations while all other species pairs were negatively correlated. The association of these two species has been noted by Brown & Curtis (1952), Whittaker (1956), McIntosh (1972), and Rogers (1978). Since hemlock canopy suppresses other species in the understory, causing low densities of subjacent saplings, gaps

formed in areas of high hemlock dominance should be more open to successful colonization by birch. Barden (1979) reported birch as the most frequent replacement of hemlock canopy trees in the Great Smoky Mountains (22 of 73 hemlock gaps observed).

Basswood is also shade-intolerant and independent saplings or seedlings were quite rare. This rarity was also noted by Daubenmire (1930) and Bray (1956) in old-growth stands of maple-basswood forests. Pigott & Huntley (1978) make the same observation regarding *Tilia cordata* in Great Britain. Basswood trees do produce abundant and vigorous basal sprouts which are densely clustered around parent trees and which appear to have a high likelihood of replacing the parent trees (Fowells, 1965). Most of the basswood canopy trees observed in this study were obviously sprouts originating from older stems. Sometimes several trees originated from a single parent stem. Cores of basswood sprouts showed growth rates several times those seen in other sapling species beneath closed canopy suggesting subsidization by the parent tree. Runkle (1979) found that basswood trees generally break off rather than uproot, a tendency which might be important in reducing mortality of sprouts upon the death of the parent tree. This literal self-replacement by basswood permits an established individual to have an extremely great life expectancy and the capability of occupying a single site for several generations of stems. Basswood canopy also appears to have some inhibitory effect on hemlock and sugar maple saplings (Fig. 8) which would encourage its self-replacement. In the plots studied beech and basswood were very rarely co-dominant at the scale sampled so no judgement of the significance of their interactions is possible.

Suggested species replacement patterns together with estimation of relative probabilities are shown in Figure 13. Additional transitions certainly occur, but those shown appear to be the most frequent. No strong directional or successional trend is suggested, though relative species importances should be expected to fluctuate over long periods of time. The probabilities suggested are likely altered somewhat by a 'mass action' effect of relative abundance of reproduction of different species. Figure 13, for instance, may be interpreted to suggest increasing importance of hemlock, but this tendency may be countered by the comparatively low abundance of

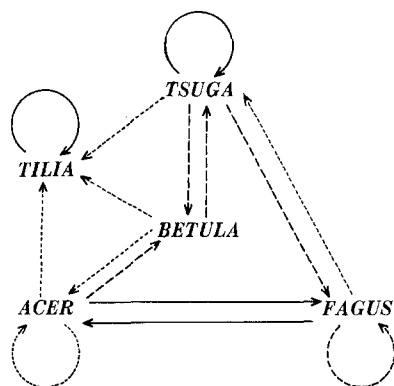


Fig. 13. Diagram showing patterns of canopy tree replacement suggested by distributions of saplings and seedlings with respect to influence of canopy species. Arrows are directed toward the species being replaced. Solid lines indicate transitions which occur most frequently, dashed lines less frequent but common transitions, and dotted lines those transitions which probably occur regularly but only occasionally. Other possible transitions almost certainly occur but more rarely.

hemlock reproduction in most forests. Maple and birch, both prolific seed producers, are often abundant as small seedlings and may be more successful than otherwise indicated. Even though hemlock saplings are usually near hemlock canopy trees, not all hemlock canopy trees have subjacent hemlock saplings and these may be replaced by other species.

The most notable and persistent deviations from the above patterns are seen in sample 9, composed of quadrats from the Tionesta stand. Here the relationship between beech and maple appears to be reversed with saplings of each species relatively more abundant near conspecific canopy trees. The relationship between beech and hemlock is also altered; saplings near canopy of both species tend to be of the opposite species. Sample 9 is also the only instance in which hemlock saplings are found at higher beech canopy influence than are beech saplings. In Tionesta maple and beech seem to be self-replacing while hemlock may replace beech with some frequency. Tionesta is the easternmost stand sampled and is widely separated from other hemlock stands studied. Geographical variation in the interactions among species is certainly possible.

The replacement patterns reported are based on present distribution patterns. If environmental changes affecting these patterns occur (and do not fluctuate around a mean with a period shorter than tree life-spans), then replacement patterns will be

altered and directional change in species composition may occur. One phenomenon which may have a strong effect on these patterns is deer browsing. Deer show strong browsing preferences among the species studied here (Hough, 1965; Moore & Johnson, 1967; Anderson & Loucks, 1979; Graham, 1954, 1958; Webb, King & Patric, 1956; Marquis, 1975). In particular, hemlock seedlings and basswood sprouts are so preferred as to be virtually eliminated in areas of high deer population. Workers differ in evaluation of effects of deer on beech, sugar maple, and yellow birch. When deer populations are extremely high these species are also browsed, but at lower deer population levels they may increase in relative importance. Recent decades have seen an increase in deer populations in most areas studied here and deer browse was noted on hemlock and basswood in most stands sampled. In a few cases reproduction by these species was virtually eliminated. If deer populations decline significantly in the near future, the decline of these tree species could be quickly reversed while extended high deer populations could have lasting impact on canopy composition.

Scale of perception and stability

Assertions of stability or instability are frequently made but are not meaningful or comparable unless accompanied by considerations of a spatial and temporal scale. Temporal scale may be defined so as to either verify or refute stability. No community is stable if the amount of time considered is great enough, while over periods much briefer than the life-spans of individual trees no significant change in community composition occurs. As the forest stands studied were judged to have been free of massive disturbance for at least several centuries, patterns observed should be largely determined by biological interactions rather than by residual effects of large disturbances.

Spatial scale is equally important. Forests exist as mosaics of patches in various stages of regeneration or change. No forest community could reasonably be called stable if samples were of such a size as to include only one or two trees as disturbance at this scale is a necessary consequence of individual mortality. Coarser scales of sampling integrate the states of many microsites occupied by individual trees. Hence, perception of stability depends on the

relative size of patches in the mosaic and sample size. At a sample size large enough to include patches of all ages, any forest type should satisfy criteria for stability. The spatial units of perception in this study are stands of a few to a few tens of hectares. At this scale population interactions should determine in the absence of massive disturbances the stability of the stand.

Explanation of patterns perceived at one scale may require consideration of phenomena occurring at other scales. Stability of forests at the scale of the stand may depend on instability at the scale of the microsite. Properties such as stability may appear at particular scales but become illusory when other scales are considered. Conflicts regarding description of community stability may often be resolved by standardization of scale.

The stability suggested for these forests is not a monolithic one involving a set of tightly co-evolved species. The modern ranges of the species studied do not coincide precisely at any scale and current species assemblages have existed only since the arrival of various species following their independent northward migration during the Holocene (Davis, 1976). Climatic change over the few millennia during which current species combinations have persisted has probably repeatedly altered community dynamics. However, whenever and wherever environmental conditions remain fairly stable for more than a generation or two, groups of tree species whose pre-existing life histories allow co-existence should become prevalent. The approximate stability suggested by the replacement patterns described must be regarded as fragile; the strongest statement that can be made is that a pattern is formed by which stable co-existence might be maintained at the defined spatial scale if the environment which produced the pattern does not and has not recently changed. Other workers (Runkle, 1981; Fox, 1977) have proposed patterns of saplings distributions and tree replacement suggesting some degree of stability in similar forests.

Forcier (1975), examining the distributions of seedlings and saplings with respect to canopy trees in a New Hampshire forest suggested a directional change involving gradual replacement of yellow birch and maple by beech. Horn (1975), in a Markov analysis of a New Jersey forest also suggested successional trends toward beech dominance. These last two results are not necessarily in conflict with

the current study, since both studies were done in stands which might be considered successional, and involved sites in the northeastern United States which was poorly sampled in this study. Results from the single eastern stand studied here (Tionesta) are potentially consistent with Forcier's study.

Conclusions

In all of the above studies canopy-understory interactions have been identified as primary forces determining both successional trends and the patterns of species co-existence in stable forests, thus emphasizing the importance of two general concerns in the study of forest communities. First, an understanding of all life cycle stages is crucial for interpretation of community structure and dynamics. The specific patterns and interactions suggested for the populations and communities studied here could not have been suggested by observations of canopy or understory patterns alone. The study of the behavior of tree species in the understory and their requirements in early life-history stages has often been neglected, with the result that understanding of niche differentiation among them is poorly developed. Further understanding of community dynamics and properties will depend on study of complete life-histories of individual species.

The second point concerns scale of perception. Interactions between life-history strategies and environmental fluctuation control community dynamics; these driving forces are universal (Denslow, 1980). Our perception of differences among communities is often due not to intrinsic differences in community mechanisms, but to variation in the spatial and temporal scales at which they function. The driving forces of succession are rooted in the modification of environment by existing species and in chance factors such as seed source and disturbance history (Drury & Nisbet, 1973; Connell & Slatyer, 1977). This study and others show that the dynamics of species interactions and patterns of tree replacement are controlled by essentially similar factors in mature, stable forests. The forces and mechanisms driving successional change and maintaining stability are not qualitatively different. Rather, similar forces acting on different parts of a continuum of spatial and temporal scales lead to communities which differ in perceived stability.

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Accepted 18.11.1983.